

1. Throughout this question the process $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with mean zero and variance σ_z^2 .

The process $\{X_t : t \in \mathbb{Z}\}$ is such that

$$X_t = Z_t - \frac{11}{12}Z_{t-1} + \frac{1}{6}Z_{t-2}.$$

- (a) Show that this process is invertible.

SOLUTION

Using the backward shift operator, B , we write the process as

$$X_t = \left(1 - \frac{11}{12}B + \frac{1}{6}B^2\right) Z_t$$

The process is invertible if the roots of

$$1 - \frac{11}{12}z + \frac{1}{6}z^2 = 0$$

lie outside the unit circle.

We see that

$$\begin{aligned} 1 - \frac{11}{12}z + \frac{1}{6}z^2 &= 0 \\ \implies \left(1 - \frac{2z}{3}\right) \left(1 - \frac{z}{4}\right) &= 0 \end{aligned}$$

and the roots are $z = 3/2$ and $z = 4$. Each root lies outside the unit circle so the process is invertible.

- (b) Determine the autocovariance generating function of $\{X_t : t \in \mathbb{Z}\}$.

SOLUTION

Consider $X_t = \alpha(B)Z_t$ where, in this example:

$$\alpha(B) = 1 - \frac{11}{12}B + \frac{1}{6}B^2.$$

The autocovariance generating function is

$$\Gamma(z) = \sigma_z^2 \alpha(z) \alpha(z^{-1})$$

Then

$$\begin{aligned} \Gamma(z) &= \sigma_z^2 \left(1 - \frac{11}{12}z + \frac{1}{6}z^2\right) \left(1 - \frac{11}{12}z^{-1} + \frac{1}{6}z^{-2}\right) \\ &= \sigma_z^2 \left(1 + \frac{121}{144} + \frac{1}{36} - \left(\frac{11}{12} + \frac{11}{72}\right)z - \left(\frac{11}{12} + \frac{11}{72}\right)z^{-1} + \frac{1}{6}z^2 + \frac{1}{6}z^{-2}\right) \\ &= \sigma_z^2 \left(\frac{269}{144} - \frac{77}{72}z - \frac{77}{72}z^{-1} + \frac{1}{6}z^2 + \frac{1}{6}z^{-2}\right). \end{aligned}$$

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- (c) Hence write down the autocovariance function of $\{X_t : t \in \mathbb{Z}\}$ in terms of σ_z^2 , explain your answer.

SOLUTION

The autocovariance at lag h is given by the coefficient of z^h in the autocovariance generating function.

Hence

$$\gamma(h) = \begin{cases} \frac{269}{144}\sigma_z^2 & h = 0; \\ -\frac{77}{72}\sigma_z^2 & |h| = 1; \\ \frac{\sigma_z^2}{6} & |h| = 2; \\ 0 & \text{otherwise.} \end{cases}$$

- (d) Determine the spectral density function of $\{X_t : t \in \mathbb{Z}\}$, expressing your answer in terms of $\cos w$ (where $w \in (-\pi, \pi]$).

SOLUTION

Using the autocovariance generating function, the spectral density function is

$$g(w) = \frac{1}{2\pi} \Gamma(e^{iw}).$$

Then

$$\begin{aligned} g(w) &= \frac{\sigma_z^2}{2\pi} \left(\frac{269}{144} - \frac{77}{72}e^{iw} - \frac{77}{72}e^{-iw} + \frac{1}{6}e^{i2w} + \frac{1}{6}e^{-i2w} \right) \\ &= \frac{\sigma_z^2}{2\pi} \left(\frac{269}{144} - \frac{77}{72}(e^{iw} + e^{-iw}) + \frac{1}{6}(e^{i2w} + e^{-i2w}) \right) \\ &= \frac{\sigma_z^2}{2\pi} \left(\frac{269}{144} - \frac{77}{36}\cos w + \frac{1}{3}\cos 2w \right) \\ &= \frac{\sigma_z^2}{2\pi} \left(\frac{269}{144} - \frac{77}{36}\cos w + \frac{1}{3}(2\cos^2 w - 1) \right) \\ &= \frac{\sigma_z^2}{2\pi} \left(\frac{2}{3}\cos^2 w - \frac{77}{36}\cos w + \frac{221}{144} \right) \end{aligned}$$

for $w \in (-\pi, \pi]$.

- (e) Sketch the spectral density function from (d) for frequency values in the range $[0, \pi]$. What does this spectral density function suggest with regard to the likely behaviour of the process $\{X_t : t \in \mathbb{Z}\}$? Explain your answer.

SOLUTION

Since

$$g(w) = \frac{\sigma_z^2}{2\pi} \left(\frac{2}{3}\cos^2 w - \frac{77}{36}\cos w + \frac{221}{144} \right)$$

we see that the spectral density is a parabola (as a function of $\cos w$).

As such, we consider the turning points where $w \in [0, \pi]$.

The first derivative of $g(w)$ is

$$g'(w) = \frac{\sigma_z^2}{2\pi} \left(-\frac{4}{3} \sin w \cos w + \frac{77}{36} \sin w \right)$$

which is equal to zero when

$$\sin w \left(\frac{77}{36} - \frac{4}{3} \cos w \right) = 0$$

which implies that $\sin w = 0$ and, hence $w = 0$ or π .

The second derivative of $g(w)$ is

$$g''(w) = \frac{\sigma_z^2}{2\pi} \left[-\frac{4}{3} (\cos^2 w - \sin^2 w) + \frac{77}{36} \cos w \right]$$

and

$$g''(0) = \frac{\sigma_z^2}{2\pi} \left(\frac{77}{36} - \frac{4}{3} \right) > 0$$

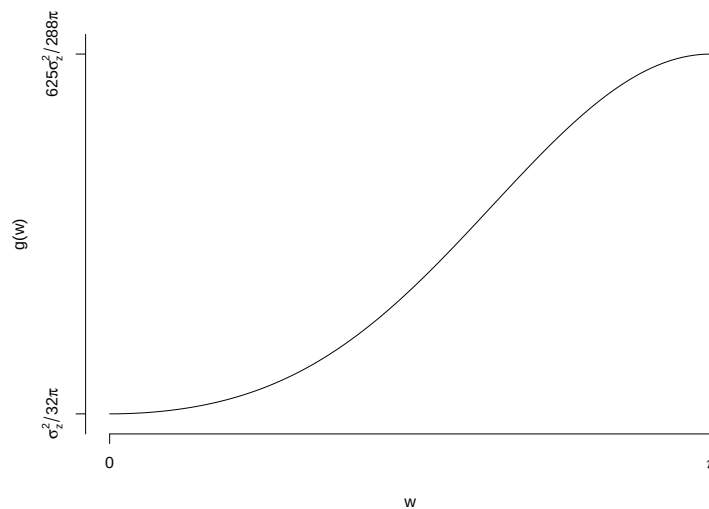
$$g''(\pi) = \frac{\sigma_z^2}{2\pi} \left(-\frac{77}{36} - \frac{4}{3} \right) < 0$$

Hence a local minimum occurs at $w = 0$ and a local maximum at $w = \pi$.

Noting that

$$g(0) = \frac{\sigma_z^2}{32\pi} \quad \text{and} \quad g(\pi) = \frac{625\sigma_z^2}{288\pi}.$$

A sketch of the spectral density function is shown below. The plot indicates that the



series will exhibit high frequency behaviour rather than longer-term trends because more of the spectral density is concentrated where the frequency is higher.

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2. Throughout this question the process $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with mean zero and variance σ_z^2 .

The (weakly) stationary process $\{X_t : t \in \mathbb{Z}\}$ is such that

$$X_t = \mu - \frac{1}{9}(X_{t-1} - \mu) + \frac{2}{27}(X_{t-2} - \mu) + Z_t$$

where μ is a real-valued parameter and $\mathbb{E}(X_t) = \mu$ for all $t \in \mathbb{Z}$.

- (a) The process $\{Y_t : t \in \mathbb{Z}\}$ is such that $Y_t = X_t - \mu$. Show that

$$\gamma_Y(h) = \gamma_X(h)$$

where $\gamma_X(h)$ and $\gamma_Y(h)$ are the autocovariance functions of $\{X_t : t \in \mathbb{Z}\}$ and $\{Y_t : t \in \mathbb{Z}\}$, respectively, with $h \in \mathbb{Z}$.

Hint: You do not need to determine $\gamma_X(h)$ or $\gamma_Y(h)$ explicitly here.

SOLUTION

Note that

$$\begin{aligned} \gamma_Y(h) &= \mathbb{E}[(Y_t - \mathbb{E}(Y_t))(Y_{t+h} - \mathbb{E}(Y_{t+h}))] \\ &= \mathbb{E}[(X_t - \mu - [\mathbb{E}(X_t - \mu)])(X_{t+h} - \mu - [\mathbb{E}(X_{t+h} - \mu)])] \\ &= \mathbb{E}[(X_t - \mu - 0)(X_{t+h} - \mu - 0)] \quad (\text{since } \mathbb{E}(X_t) = \mu \text{ for all } t) \\ &= \mathbb{E}[(X_t - \mu)(X_{t+h} - \mu)] \\ &= \gamma_X(h). \end{aligned}$$

- (b) For the process $\{Y_t : t \in \mathbb{Z}\}$, derive the Yule-Walker equations. You may assume that Y_{t-h} and Z_t are independent where $h > 0$ is a positive integer.

SOLUTION

We note that

$$Y_t = -\frac{1}{9}Y_{t-1} + \frac{2}{27}Y_{t-2} + Z_t.$$

Multiplying both sides of the above by Y_{t-h} and taking expectations we obtain

$$\begin{aligned} \mathbb{E}(Y_t Y_{t-h}) &= -\frac{1}{9}\mathbb{E}(Y_{t-1} Y_{t-h}) + \frac{2}{27}\mathbb{E}(Y_{t-2} Y_{t-h}) + \mathbb{E}(Z_t Y_{t-h}) \\ \implies \gamma_Y(-h) &= -\frac{1}{9}\gamma_Y(-h+1) + \frac{2}{27}\gamma_Y(-h+2) \end{aligned}$$

Since the process is stationary, we see that

$$\gamma_Y(h) = -\frac{1}{9}\gamma_Y(h-1) + \frac{2}{27}\gamma_Y(h-2)$$

and dividing the above by $\gamma_Y(0)$ and re-arranging yields the Yule-Walker equations

$$\rho_Y(h) + \frac{1}{9}\rho_Y(h-1) - \frac{2}{27}\rho_Y(h-2) = 0 \quad h \geq 0.$$

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- (c) By solving the Yule-Walker equations from (b), determine the autocorrelation function for the process $\{Y_t : t \in \mathbb{Z}\}$. You should show your working clearly.

SOLUTION

The auxiliary equation is

$$\begin{aligned} \lambda^2 + \frac{\lambda}{9} - \frac{2}{27} &= 0 \\ \implies \left(\lambda - \frac{2}{9}\right) \left(\lambda + \frac{1}{3}\right) &= 0 \end{aligned}$$

with roots $\lambda = \frac{2}{9}$ and $\lambda = -\frac{1}{3}$.

Hence, the general solution to the Yule-Walker equations is

$$\rho_Y(h) = A \left(\frac{2}{9}\right)^{|h|} + B \left(-\frac{1}{3}\right)^{|h|}.$$

for constants A and B .

Noting that $\rho_Y(0) = 1$:

$$A + B = 1$$

Also, for $h = 1$

$$\begin{aligned} \rho_Y(1) + \frac{1}{9}\rho_Y(0) - \frac{2}{27}\rho_Y(-1) &= 0 \\ \implies \frac{25}{27}\rho_Y(1) &= -\frac{1}{9} \\ \implies \rho_Y(1) &= -\frac{3}{25}. \end{aligned}$$

Then we solve the simultaneous equations

$$\begin{aligned} A + B &= 1 \\ \frac{2A}{9} - \frac{B}{3} &= -\frac{3}{25} \end{aligned}$$

where

$$\begin{aligned} \frac{2A}{9} - \frac{1}{3}(1 - A) &= -\frac{3}{25} \\ \implies \frac{5A}{9} &= -\frac{3}{25} + \frac{1}{3} \\ &= \frac{16}{75} \end{aligned}$$

Then $A = \frac{48}{125}$ and $B = \frac{77}{125}$.

Hence

$$\rho_Y(h) = \left(\frac{48}{125}\right) \left(\frac{2}{9}\right)^{|h|} + \left(\frac{77}{125}\right) \left(-\frac{1}{3}\right)^{|h|}.$$

TURN OVER

- (d) Show that $\mathbb{E}(Y_t Z_t) = \sigma_z^2$.

SOLUTION

Since

$$Y_t = -\frac{1}{9}Y_{t-1} + \frac{2}{27}Y_{t-2} + Z_t$$

multiplying both sides by Z_t and taking expectations, we obtain

$$\begin{aligned}\mathbb{E}(Y_t Z_t) &= -\frac{1}{9}\mathbb{E}(Y_{t-1} Z_t) + \frac{2}{27}\mathbb{E}(Y_{t-2} Z_t) + \mathbb{E}(Z_t^2) \\ &= \mathbb{E}(Z_t^2)\end{aligned}$$

Since $\mathbb{E}(Y_{t-h} Z_t) = 0$, owing to the independence of Z_t and Y_{t-h} for $h > 0$. Hence

$$\mathbb{E}(Y_t Z_t) = \mathbb{E}(Z_t^2) = \sigma_z^2 \text{ as required.}$$

- (e) A sequential sample of $\{X_t : t \in \mathbb{Z}\}$ (of size n) is observed, denoted $x_1, \dots, x_n$. Show that an approximate estimate of σ_z^2 is

$$\hat{\sigma}_z^2 = c_0 \left(1 + \frac{1}{9}r_1 - \frac{2}{27}r_2 \right)$$

where $c_0 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$ and r_j is the sample autocorrelation of the process $\{X_t : t \in \mathbb{Z}\}$ at lag j ($j \in \{1, 2\}$). Supposing $c_0 = 0.94$, $r_1 = -0.23$ and $r_2 = 0.15$, estimate σ_z^2 .

SOLUTION

Consider

$$Y_t = -\frac{1}{9}Y_{t-1} + \frac{2}{27}Y_{t-2} + Z_t.$$

Multiply both sides of the above by Y_t and take expectations to obtain

$$\begin{aligned}\mathbb{E}(Y_t^2) &= -\frac{1}{9}\mathbb{E}(Y_t Y_{t-1}) + \frac{2}{27}\mathbb{E}(Y_t Y_{t-2}) + \mathbb{E}(Y_t Z_t) \\ \implies \gamma_Y(0) &= -\frac{1}{9}\gamma_Y(1) + \frac{2}{27}\gamma_Y(2) + \sigma_z^2.\end{aligned}$$

Rearranging the above

$$\begin{aligned}\sigma_z^2 &= \gamma_Y(0) + \frac{1}{9}\gamma_Y(1) - \frac{2}{27}\gamma_Y(2) \\ &= \gamma_Y(0) \left(1 + \frac{1}{9} \frac{\gamma_Y(1)}{\gamma_Y(0)} - \frac{2}{27} \frac{\gamma_Y(2)}{\gamma_Y(0)} \right) \\ &= \gamma_Y(0) \left(1 + \frac{1}{9}\rho_Y(1) - \frac{2}{27}\rho_Y(2) \right).\end{aligned}$$

Since $\gamma_X(h) = \gamma_Y(h)$ it follows that c_0 is an estimate of $\gamma_Y(0)$ and r_1 and r_2 are estimates of $\rho_Y(1)$ and $\rho_Y(2)$, respectively. Therefore, substituting the sample values into the above, we obtain

$$\hat{\sigma}_z^2 = c_0 \left(1 + \frac{1}{9}r_1 - \frac{2}{27}r_2 \right).$$

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If $c_0 = 0.94$, $r_1 = -0.23$ and $r_2 = 0.15$ then, substituting these values into the expression from (e), we obtain

$$\begin{aligned}\hat{\sigma}_z^2 &= 0.94 \left(1 + \frac{1}{9}(-0.23) - \frac{2}{27}(0.15) \right) \\ &= 0.906.\end{aligned}$$

3. Throughout this question the process $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with mean zero and variance σ_z^2 .

The process $\{X_t : t \in \mathbb{Z}\}$ is such that

$$X_t = 0.5X_{t-1} + Z_t - 0.4Z_{t-1}.$$

- (a) What type of process is $\{X_t : t \in \mathbb{Z}\}$, including its order?

SOLUTION

This is an $ARMA(1, 1)$ process.

- (b) Is the process $\{X_t : t \in \mathbb{Z}\}$ (weakly) stationary? Justify your answer.

SOLUTION

Using the backward shift operator, we write the process as

$$(1 - 0.5B) X_t = (1 - 0.4B) Z_t.$$

The process is (weakly) stationary if the roots of

$$1 - 0.5z = 0$$

lie outside the unit circle.

Since the root is $z = 2 > 1$, the process is (weakly) stationary.

- (c) Write the process $\{X_t : t \in \mathbb{Z}\}$ as a moving average process of infinite order (i.e. in $MA(\infty)$ form). You should show your working clearly and express your final answer in its most simplified form.

SOLUTION

We have

$$(1 - 0.5B) X_t = (1 - 0.4B) Z_t.$$

Rearranging

$$X_t = (1 - 0.5B)^{-1} (1 - 0.4B) Z_t.$$

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Let $f(B) = (1 - 0.5B)^{-1}$ and consider $f^{(k)}(B)$ the k^{th} derivative of $f(B)$ with respect to B . Then

$$\begin{aligned} f^{(1)}(B) &= 0.5 (1 - 0.5B)^{-2} \\ f^{(2)}(B) &= 2(0.5)^2 (1 - 0.5B)^{-2} \\ f^{(3)}(B) &= 3!(0.5)^3 (1 - 0.5B)^{-3}. \end{aligned}$$

We see that $f^{(k)}(B) = k!(0.5)^k (1 - 0.5B)^{-(k+1)}$. The Taylor series expansion of $f(B)$ about 0 is

$$\begin{aligned} f(B) &= f(0) + f^{(1)}(0)B + \frac{f^{(2)}(0)}{2!}B^2 + \dots \\ &= \sum_{j=0}^{\infty} \frac{f^{(j)}(0)}{j!} B^j \\ &= \sum_{j=0}^{\infty} (0.5)^j B^j \end{aligned}$$

Then

$$\begin{aligned} X_t &= \sum_{j=0}^{\infty} (0.5)^j B^j (1 - 0.4B) Z_t \\ &= \sum_{j=0}^{\infty} 0.5^j Z_{t-j} - 0.4 \sum_{j=0}^{\infty} 0.5^j Z_{t-j-1} \\ &= Z_t + \sum_{j=1}^{\infty} (0.5^j - 0.4(0.5^{j-1})) Z_{t-j} \\ &= Z_t + 0.1 \sum_{j=1}^{\infty} 0.5^{j-1} Z_{t-j} \end{aligned}$$

- (d) Using your answer to (c), determine the autocorrelation function of $\{X_t : t \in \mathbb{Z}\}$.

SOLUTION

Suppose $h > 0$ then

$$\begin{aligned} \text{Cov}(X_t, X_{t+h}) &= \mathbb{E}(Y_t Y_{t+h}) - \mathbb{E}(Y_t) \mathbb{E}(Y_{t+h}) \\ &= \mathbb{E}(Y_t Y_{t+h}) \\ &= \mathbb{E} \left(\sum_{j=0}^{\infty} \psi_j Z_{t-j} \sum_{k=0}^{\infty} \psi_k Z_{t+h-k} \right) \\ &= \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} \psi_j \psi_k \mathbb{E}(Z_{t-j} Z_{t+h-k}). \end{aligned}$$

Since $\{Z_t\}$ is a white noise process, $\mathbb{E}(Z_{t-j} Z_{t+h-k}) = 0$ except where $h - k = -j$. In that case

$$k = j + h$$

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and it follows that

$$\begin{aligned}\text{Cov}(X_t, X_{t+h}) &= \sum_{j=0}^{\infty} \psi_j \psi_{j+h} \mathbb{E}(Z_{t-j}^2) \\ &= \sigma_z^2 \sum_{j=0}^{\infty} \psi_j \psi_{j+h}\end{aligned}$$

Now

$$\psi_j = \begin{cases} 1 & j = 0; \\ 0.1(0.5^{j-1}) & j > 0. \end{cases}$$

Then for $h \geq 0$

$$\begin{aligned}\text{Cov}(X_t, X_{t+h}) &= \sigma_z^2 \sum_{j=0}^{\infty} \psi_j \psi_{j+h} \\ &= \sigma_z^2 \left(\psi_h + \sum_{j=1}^{\infty} [0.1(0.5)^{j-1}] [0.1(0.5)^{j+h-1}] \right) \\ &= \sigma_z^2 \left(\psi_h + (0.1^2)(0.5^h) \sum_{j=1}^{\infty} (0.5^2)^{j-1} \right) \\ &= \sigma_z^2 \left(\psi_h + 0.01(0.5^h) \sum_{j=1}^{\infty} (0.25)^{j-1} \right) \\ &= \sigma_z^2 \left(\psi_h + 0.01(0.5^h) \frac{1}{1 - 0.25} \right) \\ &= \sigma_z^2 \left(\psi_h + \frac{0.01(0.5^h)}{0.75} \right)\end{aligned}$$

So for $h = 0$

$$\begin{aligned}\text{Var}(X_t) &= \sigma_z^2 \left(1 + \frac{0.01(0.5^h)}{0.75} \right) \\ &= \frac{76\sigma_z^2}{75}.\end{aligned}$$

For $h > 0$

$$\begin{aligned}\text{Cov}(X_t, X_{t+h}) &= \sigma_z^2 \left(0.1(0.5^{h-1}) + \frac{0.01(0.5^h)}{0.75} \right) \\ &= \sigma_z^2 \left(\frac{0.1}{0.5} + \frac{0.01}{0.75} \right) 0.5^h \\ &= \sigma_z^2 \frac{16(0.5^h)}{75}.\end{aligned}$$

Since the process is stationary, by symmetry

$$\text{Cov}(X_t, X_{t-h}) = \sigma_z^2 \frac{16(0.5^{|h|})}{75}.$$

TURN OVER

Thus the autocovariance function is

$$\gamma_X(h) = \begin{cases} \frac{76\sigma_z^2}{75} & h = 0; \\ \sigma_z^2 \frac{16(0.5^{|h|})}{75} & h = \pm 1, \pm 2, \dots \end{cases}$$

The autocorrelation function is

$$\rho_X(h) = \begin{cases} 1 & h = 0; \\ \frac{16}{76}(0.5^{|h|}) & h = \pm 1, \pm 2, \dots \end{cases}$$

- (e) A large sequential sample of $\{X_t : t \in \mathbb{Z}\}$ is taken but the following model is fitted to the observed data:

$$X_t = 0.5X_{t-1} + Z_t.$$

The residuals from this fitted model are calculated and a plot of the sample autocorrelation function of these residuals is produced for lags 0 to 15. Describe how you would expect this plot to look. Explain your answer.

SOLUTION

The true process is

$$\begin{aligned} X_t &= 0.5X_{t-1} + Z_t - 0.4Z_{t-1} \implies (1 - 0.5B)X_t = (1 - 0.4B)Z_t \\ \implies X_t &= (1 - 0.5B)^{-1}(1 - 0.4B)Z_t \end{aligned}$$

The residuals from the fitted model are

$$\hat{Z}_t = (1 - 0.5B)X_t$$

Using the form of the true process, this implies that

$$\begin{aligned} \hat{Z}_t &= (1 - 0.5B)(1 - 0.5B)^{-1}(1 - 0.4B)Z_t \\ &= (1 - 0.4B)Z_t \\ &= Z_t - 0.4Z_{t-1}. \end{aligned}$$

We see that the residuals $\hat{Z}_t$ are an MA(1) process.

A plot of the ACF against the lag would show a negative spike at lag 1 and then cut off to zero.

4. Throughout this question the process $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with mean zero and variance σ_z^2 .

The process $\{X_t : t \in \mathbb{Z}\}$ is an MA(1) process, written

$$X_t = Z_t + \theta Z_{t-1}$$

with $|\theta| < 1$.

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- (a) Determine the autocorrelation function of $\{X_t : t \in \mathbb{Z}\}$. You should show your working clearly.

SOLUTION

Consider $\gamma_X(h)$ for $h > 0$:

$$\begin{aligned}\gamma_X(h) &= \text{Cov}(X_t, X_{t+h}) \\ &= \mathbb{E}(X_t X_{t+h}) \\ &= \mathbb{E}[(Z_t + \theta Z_{t-1})(Z_{t+h} + \theta Z_{t+h-1})] \\ &= \mathbb{E}(Z_t Z_{t+h}) + \theta [\mathbb{E}(Z_{t-1} Z_{t+h-1}) + \mathbb{E}(Z_{t-1} Z_{t+h})] + \theta^2 \mathbb{E}(Z_{t-1} Z_{t+h-1})\end{aligned}$$

For $h = 0$:

$$\begin{aligned}\gamma_X(0) &= \mathbb{E}(Z_t^2) + \theta^2 \mathbb{E}(Z_{t-1}^2) \\ &= \sigma_z^2(1 + \theta^2)\end{aligned}$$

For $h = \pm 1$

$$\gamma_X(h) = \sigma_z^2 \theta$$

and $\gamma_X(h) = 0$ for $|h| > 1$.

The autocorrelation function is

$$\rho_X(h) = \begin{cases} 1 & h = 0; \\ \frac{\theta}{1+\theta^2} & |h| = \pm 1; \\ 0 & |h| \geq 2. \end{cases}$$

- (b) A sequential sample of $\{X_t : t \in \mathbb{Z}\}$ of size n is recorded, denoted $x_1, \dots, x_n$. Suppose that $\bar{x}$ is the sample mean, $c_0 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$ and r_1 is the sample autocorrelation at lag 1. Determine method of moments estimates of θ and σ_z^2 in terms of c_0 and r_1 .

SOLUTION

We equate the sample and theoretical moments:

$$\begin{aligned}\frac{\hat{\theta}}{1 + \hat{\theta}^2} &= r_1 \\ \implies \hat{\theta} &= r_1(1 + \hat{\theta}^2) \\ \implies r_1 \hat{\theta}^2 - \hat{\theta} + r_1 &= 0 \\ \implies \hat{\theta} &= \frac{1 \pm \sqrt{1 - 4r_1^2}}{2r_1}\end{aligned}$$

The chosen estimate of $\hat{\theta}$ would be that which ensures the process is invertible (i.e. $|\hat{\theta}| < 1$).

We know that

$$\text{Var}(X_t) = \sigma_z^2(1 + \theta^2).$$

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To estimate σ_z^2 we set

$$\begin{aligned}\hat{\sigma}_z^2(1 + \hat{\theta}^2) &= c_0 \\ \implies \hat{\sigma}_z^2 &= \frac{c_0}{1 + \hat{\theta}^2}.\end{aligned}$$

(c) Write the process $\{Z_t : t \in \mathbb{Z}\}$ in the form

$$Z_t = \sum_{j=0}^{\infty} a_j X_{t-j}$$

where the coefficients a_j are real-valued functions of θ . You should show your working clearly.

SOLUTION

We write the process as

$$\begin{aligned}X_t &= (1 + \theta B)Z_t \\ \implies Z_t &= (1 + \theta B)^{-1}X_t\end{aligned}$$

Let $g(B) = (1 + \theta B)^{-1}$ then if $g^{(k)}(B)$ is the k^{th} derivative of $g(B)$:

$$\begin{aligned}g^{(1)}(B) &= -\theta(1 + \theta B)^{-2} \\ g^{(2)}(B) &= (-1)^2 2\theta^2(1 + \theta B)^{-3} \\ g^{(3)}(B) &= (-1)^3 3!\theta^3(1 + \theta B)^{-4}\end{aligned}$$

In general $g^{(k)}(B) = (-1)^k k! \theta^k (1 + \theta B)^{-(k+1)}$ and the Taylor expansion of $g(B)$ about zero is

$$\begin{aligned}g(B) &= \sum_{k=0}^{\infty} \frac{g^{(k)}(0)}{k!} B^k \\ &= \sum_{k=0}^{\infty} (-1)^k \theta^k B^k.\end{aligned}$$

Then

$$Z_t = \sum_{k=0}^{\infty} (-1)^k \theta^k X_{t-k}.$$

(d) The process $\{Y_t : t \in \mathbb{Z}\}$ measures the annual rainfall in millimetres (mm) at a weather station and it is assumed that

$$Y_t = \mu + Z_t + \psi Z_{t-1}$$

where μ and ψ are real-valued parameters. Suppose that we have 30 years of observations of this process from 1994 to 2023, denoted $y_{1994}, \dots, y_{2023}$. The sample

CONTINUED

mean of these observations is $\bar{y} = 705.6$ and the sample autocorrelation at lag 1 is $r_1 = 0.48$. In addition, $\frac{1}{30} \sum_{i=1994}^{2023} (y_i - \bar{y})^2 = 192.1$.

Calculate the method of moments estimates of the parameters μ , ψ and σ_z^2 .

SOLUTION

To estimate μ we set

$$\hat{\mu} = \bar{y} = 705.6$$

Using the answer to (b), we equate the sample and theoretical autocorrelation at lag 1.

$$\hat{\psi} = \frac{1 \pm \sqrt{1 - 4 \times 0.48^2}}{2 \times 0.48}$$

and $\hat{\psi} = \frac{4}{3}$ or $\hat{\psi} = \frac{3}{4}$. We choose $\hat{\psi} = \frac{3}{4}$.

The estimate of σ_z^2 is

$$\begin{aligned} \hat{\sigma}_z^2 &= \frac{c_0}{1 + \hat{\psi}^2} \\ &= \frac{192.1}{1 + 0.75^2} \\ &= 122.94. \end{aligned}$$

- (e) The observed values of the process $\{Y_t : t \in \mathbb{Z}\}$ (described in (d)) for the years 2019–2023 are shown in the table below. Using the above information, together

Year	2019	2020	2021	2022	2023
Annual rainfall (mm)	705	722	685	682	701

with your parameter estimates from (d), provide an optimal prediction for the 2024 annual rainfall. Here, the term ‘optimal’ implies that the predicted value is that which minimises the mean-square prediction error at a given time point. Note: If you did not obtain an answer to (d) you may assume that $\hat{\mu} = 704$, $\hat{\psi} = 0.8$ and $\hat{\sigma}_z^2 = 120$ (these are not the correct answers to (d)).

SOLUTION

Suppose that the series is observed to time n . We define

$$[Z_{n+k}] = \begin{cases} 0 & \text{if } k > 0; \\ Z_{n+k} & \text{if } k \leq 0. \end{cases}$$

Set

$$Y_t = X_t - \mu = Z_t + \psi Z_{t-1}.$$

Then

$$\hat{Y}_{n+k} = [Z_{n+k}] + \psi[Z_{n+k-1}].$$

TURN OVER

For $n = 2023$ and $k = 1$:

$$\begin{aligned}
 \hat{Y}_{2024} &= [Z_{2024}] + \psi[Z_{2023}] \\
 &= \psi Z_{2023} \\
 &= \psi \sum_{k=0}^{\infty} (-1)^k \psi^k Y_{2023-k} \\
 &= \sum_{k=0}^{\infty} (-1)^k \psi^{k+1} Y_{2023-k}
 \end{aligned}$$

Using the data and the estimate $\hat{\psi} = 0.75$:

$$\begin{aligned}
 \hat{Y}_{2024} &= 0.75Y_{2023} - 0.75^2Y_{2022} + 0.75^3Y_{2021} - 0.75^4Y_{2020} + 0.75^5Y_{2019} \\
 &= 0.75 \times (701 - \hat{\mu}) - 0.75^2 \times (682 - \hat{\mu}) + 0.75^3 \times (685 - \hat{\mu}) \\
 &\quad - 0.75^4(722 - \hat{\mu}) + 0.75^5(705 - \hat{\mu}) \\
 &= 0.75 \times (701 - 705.6) - 0.75^2 \times (682 - 705.6) + 0.75^3 \times (685 - 705.6) \\
 &\quad - 0.75^4(722 - 705.6) + 0.75^5(705 - 705.6) \\
 &= -4.19707.
 \end{aligned}$$

To transform this back to $\hat{X}_{2024}$ we set

$$\begin{aligned}
 \hat{X}_{2024} &= \hat{Y}_{2024} + \hat{\mu} \\
 &= -4.19707 + 705.6 \\
 &= 701.40.
 \end{aligned}$$